

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI**



**A Mini Project Work Report on
“LPG Gas Leakage Detection Using Arduino”**

Submitted in partial fulfilment of the requirements for the award of the degree of

**Bachelor of Engineering
In
Electronics and Telecommunication Engineering**

Submitted By

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CERTIFICATE

This is to certify that the project work entitled "**LPG Gas Leakage Detection Using Arduino**" is carried out by **Darshan N J (4JN23ET013)**, **Indudhara K (4JN23ET020)**, **Manoj Kumar G G (4JN23ET028)**, **Rohith Shreyams Chulaki (4JN23ET045)**, the bona fide students of JNN College of Engineering, Shimoga, in partial fulfilment of the requirements for the award of "Bachelor of Engineering" in the Department of "Electronics and Telecommunication Engineering of the Visvesvaraya Technological University, Belagavi, during the year 2025-2026. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved, as it satisfies the academic requirements for the project work prescribed for the said degree.

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ABSTRACT

LPG (Liquefied Petroleum Gas) is one of the most commonly used fuels in domestic kitchens, hotels, laboratories, and small-scale industries due to its high efficiency and clean combustion. However, LPG is highly flammable, and even a minor leakage can lead to severe fire accidents, explosions, and health hazards. Therefore, early detection of gas leakage is essential to prevent life-threatening situations and property damage. This project presents an Arduino-based LPG gas leakage detection and alert system designed to continuously monitor the presence of LPG in the surrounding environment. The core sensing component used is the MQ-2 gas sensor, which is capable of detecting LPG, methane, and other combustible gases with high sensitivity. The analogue output from the sensor is processed by the Arduino Uno, which compares the gas concentration with a predefined safety threshold. When leakage is detected, the system immediately activates safety measures, including sounding a buzzer alarm, turning on LED indicators, and displaying a warning message on the LCD. To further enhance safety, an exhaust fan is automatically switched on to expel the accumulated gas from the area. Additionally, a GSM module is integrated to send SMS alerts to the user, ensuring remote monitoring and timely action even when no one is present near the system. The entire setup operates on a low-power microcontroller and uses easily available electronic components, making it cost-effective and suitable for home installations. The system is simple to implement, reliable, and provides real-time detection with quick response. This project demonstrates a practical and efficient solution to improve household and industrial safety by preventing accidents caused by LPG leakage.

Keywords: LPG detection, MQ-6 sensor, Arduino, safety system, gas leakage alert.

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Introduction

1.1 Introduction

Liquefied Petroleum Gas (LPG) has become an essential energy source across domestic, commercial, and industrial sectors due to its high efficiency, ease of availability, and affordable cost. It is primarily used for cooking, heating, and certain industrial applications. Although LPG is a convenient and clean fuel, it is also highly hazardous. LPG is heavier than air and tends to accumulate near the ground when leaked, forming explosive mixtures with air. Even a small spark can ignite leaked LPG and cause severe fires or explosions.

Many LPG-related accidents occur each year due to gas leakage caused by faulty equipment, such as worn-out tubes, damaged regulators, improper stove connections, or negligence, such as leaving the burner knob open. Traditional methods for detecting gas leaks, like manually smelling gas or checking with soap and water, are not reliable, especially when gas accumulates unnoticed. For this reason, there is a critical need for an automated system that can detect leakage early and provide immediate warnings.

This project proposes a **microcontroller-based LPG Gas Leakage Detection System using Arduino, MQ-6 Gas Sensor, and GSM Module**, designed to detect leakage at the earliest possible stage. When the MQ-6 sensor senses an abnormally high concentration of LPG in the air, the Arduino processes this signal and activates an alarm system consisting of a buzzer and LED indicator. Additionally, the GSM module sends an SMS alert to the registered mobile number, notifying the user remotely. This multi-alert approach makes the system highly effective, especially when no one is present at the site of leakage.

The proposed system is cost-effective, compact, easy to assemble, and suitable for both households and commercial establishments. The combination of real-time monitoring, audible/visual alerts, and mobile communication significantly enhances safety and reduces the chances of devastating accidents.

1.2 Literature Survey

A **literature survey** is a structured review of previously published work related to your project, topic, or research area. It involves collecting information from books, research papers, journals, websites, and technical reports to understand the existing technologies, methods, and findings. The purpose of a literature survey is to analyse and compare earlier work, identify strengths and limitations, and highlight the need for your proposed project. It shows that you have good background knowledge and that your project is built on previous studies.

- **A Gas Leakage Detector Using IOT** — by C K Gomathy, S. Megha Shyam Raju, K. Sheshank Reddy. Published in International Research Journal of Engineering and Technology (IRJET), Vol. 8, Issue 10, Oct 2021.[1] The authors conclude that their proposed system — using a gas sensor, microcontroller (Arduino), IoT connectivity, buzzer alarm + LCD (and exhaust fan) — offers a cost-efficient and effective solution to detect and respond to combustible gas leaks (such as LPG, CNG) in residential or industrial settings. They claim the device can continuously monitor surroundings, detect gas leakage under most atmospheric conditions, alert users immediately (via alarm + display), and activate ventilation (exhaust fan) to disperse leaked gas — thereby minimising risk of fire or explosion. The system, with small modifications, can also be adapted for household use, not just industrial environments. Given the widespread use of LPG/gas in households and the serious safety hazards of leaks, the authors argue that this project could be a “boom” for safety — improving security in gas usage across homes and industries.
- **Implementation and design gas leakage detection system using an ATmega8 microcontroller** — by N. Evalina and H. A. Azis. Published in IOP Conference Series: Materials Science and Engineering, Vol. 821, No. 1, 2020.[2] The implemented system — using Arduino UNO along with an MQ5 Gas Sensor (plus LCD, buzzer, relay and GSM-module) — successfully detects the presence of LPG in the surrounding air, and triggers both local and remote alerts when gas concentration exceeds a preset threshold. The dual-alert mechanism (instant audible alarm + SMS notification to external contacts) ensures timely warning to occupants as well as responsible persons who may be away, reducing the risk of accidents or fire due to undetected leakage. By additionally using a relay to cut off the electrical supply upon leak detection, the design adds a safety control measure — preventing ignition sources from interacting with leaked LPG. Overall, the project demonstrates a cost-effective, practical, and easily deployable detection-and-alert system suitable for both domestic and industrial settings. If widely adopted, it can significantly lower the likelihood of gas-leakage accidents, protect lives and property, and bring greater peace of mind.

- **Gas leakage detection and smart alerting and prediction using IoT** — by Asmitha Varma, Prabhakar S, and Kayalvizhi Jayavel. Appears cited in a review list of LPG/gas-leakage detection papers.[3] Aerial pollution is a significant problem nowadays that is focusing attention on the medical and healthcare sector. The level of air pollution has been worse over time due to a variety of factors, including population growth, excessive car usage, industry, & urbanization, which have negative repercussions on human eudaimonia by directly affecting public health. The security, health, and safety of the people depend critically on the quality of the air. The degradation of the air quality in many Indian cities is one offender. The main cause of pollution is particulate matter (PM 2.5), which harms people's health by causing disorders of the respiratory system or various metabolic processes. Therefore, cheap pollution monitoring equipment that can be easily discarded anywhere is required. It has been consistently shown that those who breathe airborne particles are more susceptible to developing cancer than nonsmokers who are exposed to secondhand smoke. This paper suggests a low-cost multimodality sensor-based system as a remedy, with MQ 2 being used for sensing the presence of propane, hydrogen, and LPG (liquefied petroleum gas), MQ 5 and MQ 9 being used in gas leakage appliances to check leakage, MQ 6 being used for checking the contamination of additional gases present in the air, MQ 135 being used for air quality parameter sensor, and DHT 11 being used for measuring the temperature and humidity of the atmosphere for better quality. The user will be able to see the data from these sensors on the 24*4 LCD (Liquid Crystal Display) when they interface with the Arduino UNO microcontroller and transfer it to the cloud. The user may thus have a clear understanding of the air quality in his immediate surroundings.
- L. Jinfeng, C. Chen, C. Xiaowei and W. Wei, "**Management of Indoor Gas Safety based on the NB-IoT Gas Meter,**" 2021 33rd Chinese Control and Decision Conference (CCDC), Kunming, China, 2021.[4] Natural and human sources are contributing enormously to the rise of air pollution. The negative impacts of air pollution extend beyond people and other living things, including plants, animals, and even historical sites. As a result, society now must do an essential task: air monitoring. A portable model for remote web-based monitoring of carbon monoxide (CO) and carbon dioxide (CO₂) gases in outdoor areas is part of the Atmospheric Air Surveil System (AASS). A microprocessor, gas sensors, and a Global Positioning System (GPS) make up the sensor mote (device). The microcontroller is utilised to process the measurements of the surrounding gas concentrations made using the calibrated sensor grid. Through the Internet of Things (IoT), the motes' processed data is sent to the cloud. These data are transmitted to the server via the MQTT (Message Queuing Telemetry

Transport) protocol, where they are then stored in the Data Acquisition (DAQ) unit. The dashboard is available to display the data that has been saved. Functionalities include email alerts, activity logs, geographic location verification (GLV), and device authentication (DA). The scalability of this system has been evaluated. Additionally, the authorised user can operate the entire gadget from a place of their choosing remotely. This system is being evaluated for its capacity to scale.

- Sarika K. Shinde & Priya R. Khore **Automatic LPG Cylinder Booking and Leakage Detection System.** [5] The developed LPG monitoring system effectively integrates gas leakage detection, automatic cylinder weight measurement, and GSM-based alerting using the Arduino platform. The gas sensor accurately identifies leakage and immediately activates safety measures such as buzzer alarms and exhaust operation, ensuring a quick response to hazardous situations. The load-cell-based weight monitoring enables timely notification for cylinder refilling, reducing user inconvenience and preventing sudden unavailability of LPG. Overall, the system provides a low-cost, reliable, and user-friendly solution that enhances both safety and convenience in domestic LPG usage.

1.3 Aim of the Project

This project aims to design and develop an IoT-based LPG gas leakage detection system that continuously monitors gas levels and identifies hazardous leakage in real time. The system aims to alert users immediately through alarms and IoT notifications while activating safety measures like an exhaust fan. It focuses on enhancing household and industrial safety using a low-cost, reliable, and automated gas detection mechanism.

1.4 Objectives

1. To detect the presence of LPG gas leakage and activate a buzzer.
2. To display the gas leakage status on the LCD and automatically turn on the exhaust fan to reduce the gas concentration.
3. To send an alert message to the user's mobile phone using a GSM module, and the servo motor automatically turns off the gas regulator when gas leakage is detected.
4. To ensure safety in domestic and industrial areas by using this system.

1.5 Methodology

1. Methodology for Detecting LPG Leakage and Activating a Buzzer.

The detection mechanism starts with placing the MQ-series gas sensor near the LPG storage or usage area so that even the smallest leakage can be detected early. The sensor continuously samples the surrounding air and generates an analogue voltage proportional to the LPG concentration. This analogue value is fed into the Arduino's analogue input pin, where it is converted into a digital value by the ADC. The system is programmed with a calibrated threshold based on safe and unsafe gas levels to ensure reliable measurement. Calibration is performed by exposing the sensor to clean air and known concentrations of LPG to achieve accurate baseline readings. The Arduino uses continuous monitoring logic that checks the sensor values at very short time intervals to avoid delays in detection. When the measured concentration exceeds the safe limit, the system identifies the scenario as a gas leakage. At this point, the Arduino immediately activates a high-pitch buzzer, providing an instant audible alert. The buzzer remains active as long as the sensor detects dangerous levels of gas, ensuring constant awareness for anyone nearby. This method ensures rapid recognition of potentially hazardous situations. Additionally, the buzzer sound level is chosen so it can be heard across multiple rooms, improving safety coverage. The methodology provides a strong first line of defence by combining precise sensing with immediate alert generation, minimising the chances of unnoticed leakage.

2. Methodology for Displaying Gas Status and Automatically Turning ON the Exhaust Fan.

After processing the sensor data, the Arduino displays the gas concentration and safety status on a 16×2 LCD module, offering real-time updates to the user. The LCD continuously refreshes its readings, giving clear visual indicators such as "Safe," "Moderate," "Warning," or "Leak Detected." These dynamic messages help users quickly understand the environmental condition without needing technical knowledge. When the system detects leakage conditions, the Arduino activates the exhaust fan using a relay driver module that ensures safe switching of AC loads. The exhaust fan is placed strategically to expel the leaked gas from the area and promote ventilation. This mechanism immediately reduces LPG concentration by replacing contaminated air with fresh air from outside. The exhaust fan continues to run until the gas concentration drops to a safe level, preventing further risk. During this time, the LCD provides constant updates to help users monitor recovery progress. The fan's operation is automated, eliminating the need for manual switching in hazardous situations. This methodology significantly reduces the possibility of ignition, as LPG vapours are quickly dispersed. The integration of display, detection, and

ventilation ensures a multi-layered approach to hazard reduction. Overall, the system gives users continuous visualisation of the gas levels and ensures active ventilation whenever needed.

3. Methodology for Sending SMS Alerts and Automatically Turning OFF the Gas Regulator.

For remote safety, a GSM module such as SIM800L or SIM900 is interfaced with the Arduino through serial communication. When the gas concentration exceeds the safe limit, the Arduino prepares a structured message containing an emergency alert about the leakage. This message is instantly sent to the user's mobile phone, ensuring that they are informed even if they are away from the premises. The GSM module operates through standard mobile networks, making it a reliable method for long-distance alerts. Alongside this, the Arduino simultaneously triggers a servo motor fixed to the gas regulator via a mechanical attachment. The servo is programmed to rotate to a precise angle that closes the regulator fully, preventing additional LPG from escaping. This automatic mechanism eliminates the need for human intervention during emergencies, especially when quick action is required. The system may also be configured to send repeated SMS alerts until the leakage is handled or the environment becomes safe. This dual safety mechanism of remote alerting and mechanical shutoff significantly improves the reliability of the system. It ensures the user can take action immediately or inform emergency services if they are far away. This methodology greatly enhances the safety scope by ensuring gas supply control and communication even in the absence of human supervision.

4. Methodology for Ensuring Safety in Domestic and Industrial Areas.

The LPG detection system is designed to operate continuously, offering real-time monitoring for homes, restaurants, laboratories, hotels, and small industries that rely on LPG. All system components are carefully integrated to function as a unified safety solution. The MQ sensor continuously scans the environment for gas concentration changes, while the buzzer and LCD ensure local notification. The exhaust fan automatically improves ventilation to reduce hazardous gas accumulation. The GSM module provides remote alerts to users, and the servo motor ensures that the gas supply is shut off during emergencies. This coordination ensures that multiple layers of safety are active simultaneously. The system uses low-cost, energy-efficient components, enabling long-term use with minimal maintenance. The modular structure allows easy upgrades, such as integrating IoT modules, Wi-Fi connectivity, or cloud-based monitoring. This makes the system scalable and adaptable to both small and large environments. The design ensures that the system can function even without constant human monitoring, making it suitable for 24/7 safety applications. By combining detection, alerting, ventilation, communication, and automatic shutoff,

this methodology creates a highly reliable safety ecosystem. It greatly reduces risks of fire, explosion, and property loss, thus improving overall safety in domestic and industrial LPG applications.

1.6 Scope of the Project

The scope of this project extends toward designing and developing a low-cost, efficient, and reliable LPG gas leakage detection and safety management system using Arduino technology. It emphasises the use of an MQ-series gas sensor for accurate detection of LPG concentration and integrates multiple safety components such as buzzers, exhaust fans, GSM modules, and an automatic regulator shut-off mechanism. The project aims to create a compact and user-friendly system that can be installed in households, hotels, restaurants, laboratories, and small industrial environments where LPG is used frequently.

The project covers real-time monitoring of gas levels, enabling continuous detection through the sensor to provide immediate awareness of leakage. It includes the automatic activation of buzzers to deliver instant audible alerts, ensuring people nearby can react promptly. The scope also involves the automatic switching of the exhaust fan to reduce gas concentration effectively and improve ventilation. A GSM module is incorporated to send emergency SMS alerts so that users are informed even when they are away from the premises.

Furthermore, the project demonstrates how microcontroller-based systems can significantly improve safety by automating crucial actions such as shutting off the gas regulator using a servo motor. It highlights the importance of integrating hardware and software components to build a holistic safety system. The scope extends to ensuring that the system is easy to install, simple for users to operate, and cost-effective enough for widespread adoption.

Additionally, the project explores the adaptability of the system for future enhancements, such as IoT-based monitoring through Wi-Fi or cloud platforms. This opens possibilities for remote dashboards, mobile app integration, data logging, and predictive alerts. The system's modular structure allows its components to be upgraded or replaced without redesigning the entire setup. The project also focuses on creating a scalable solution that can be expanded for large kitchens or industrial setups.

In summary, the scope not only includes detecting and alerting users about LPG leakage but also emphasises automation, safety enhancement, user convenience, and technological upgradability. It demonstrates the potential of low-cost embedded systems in reducing fire hazards and protecting lives and property in LPG-based environments.

1.7 Limitations

1. Limited Sensor Accuracy

MQ-series sensors have moderate accuracy and are highly affected by humidity, temperature, dust, and airflow. They often produce fluctuating readings that may cause false alarms or missed leakages. These sensors also detect other gases like smoke and alcohol vapours, which reduces their specificity for LPG. Over time, sensor drift occurs, making long-term accuracy unreliable.

2. Calibration Requirement

The MQ sensor requires frequent calibration to maintain proper sensitivity, but most users do not perform it regularly. Incorrect or missing calibration can result in false readings or delayed detection. Environmental factors like dust, moisture, and temperature affect calibration, making it difficult to maintain in real-world installations.

3. Short Sensor Lifespan

Continuous heating elements inside MQ sensors cause them to degrade over time, reducing detection efficiency. The sensing material becomes less responsive after months of use, requiring replacement. Exposure to dust, oil fumes, and moisture shortens the lifespan further, making long-term reliability a challenge.

4. Power Supply Sensitivity

The system requires a stable power supply; voltage drops or fluctuations can reset the Arduino or GSM module. GSM modules need higher current, and any power instability may prevent sending SMS alerts. In areas with frequent power cuts, the system may fail without a dedicated backup source.

5. Limited Communication Range

The GSM module's alert system depends entirely on mobile network availability. Poor signal strength or network outages can delay or block SMS alerts. The system may also take time to reconnect to the network after restarts, reducing the reliability of communication during emergencies.

6. No Automatic Shutoff in Basic System

Low-cost versions of the system only detect leakage and provide alerts, but cannot automatically turn off the gas supply. Without a servo or solenoid valve, leakage may continue even after detection, increasing hazard risks. Adding an automatic shutoff mechanism increases cost and complexity.

7. Cannot Detect All Gases

MQ-2 sensors are sensitive to LPG but also respond to other gases like methane, alcohol, and smoke. This multi-gas sensitivity may trigger unwanted alarms. The system is not suitable for detecting industrial hazardous gases that require specialised sensors.

8. Sensor Warm-up Delay

MQ sensors require 1–2 minutes of heating time to stabilise before giving accurate readings. During this warm-up period, the system cannot detect leakage effectively. This delay reduces immediate protection when the device is first powered on.

9. Environmental Limitations

Extreme temperatures, dust, and moisture significantly affect sensor performance. Kitchens with heavy oil fumes may reduce sensor sensitivity over time. This makes the system less suitable for harsh or unclean environments without regular maintenance.

10. Not Fail-Safe

If any component, like the sensor, GSM module, or wiring, fails, the system may stop working without giving a warning. The Arduino does not have built-in self-diagnosis to detect internal failures. This lack of fail-safe mechanisms reduces overall reliability in critical situations.

1.8 Organisation of the project

This project report is organised into several chapters. **Chapter 1** presents the introduction, objectives, problem statement, and scope of the project. **Chapter 2** covers the design and implementation of the project, including the block diagram and flowchart. **Chapter 3** presents the results of the project along with the discussion. **Chapter 4** concludes with the future scope of the project

Design and implementation

2.1 Introduction

This chapter focuses on the design and implementation of the LPG Gas Leakage Detection System, beginning with the explanation of the block diagram and flowchart. The block diagram provides a clear overview of the system architecture, highlighting key components such as the MQ gas sensor, Arduino microcontroller, buzzer, exhaust fan, GSM module, LCD, and automatic regulator shut-off mechanism. The flowchart outlines the step-by-step logical operations of the system, from gas detection to alert generation and safety actions. Following this, the chapter describes the hardware and software requirements essential for building the system, including the selection of sensors, communication modules, actuators, and the Arduino programming used to control the sensing and alert mechanisms. The implementation section explains the integration of all components, detailing how the sensor readings are processed, how alerts are triggered, and how automatic safety responses such as fan activation and regulator closure are executed. The chapter concludes with a summary of the challenges faced during implementation and the solutions adopted to ensure reliable, efficient, and safe system operation.

2.2 Block Diagram

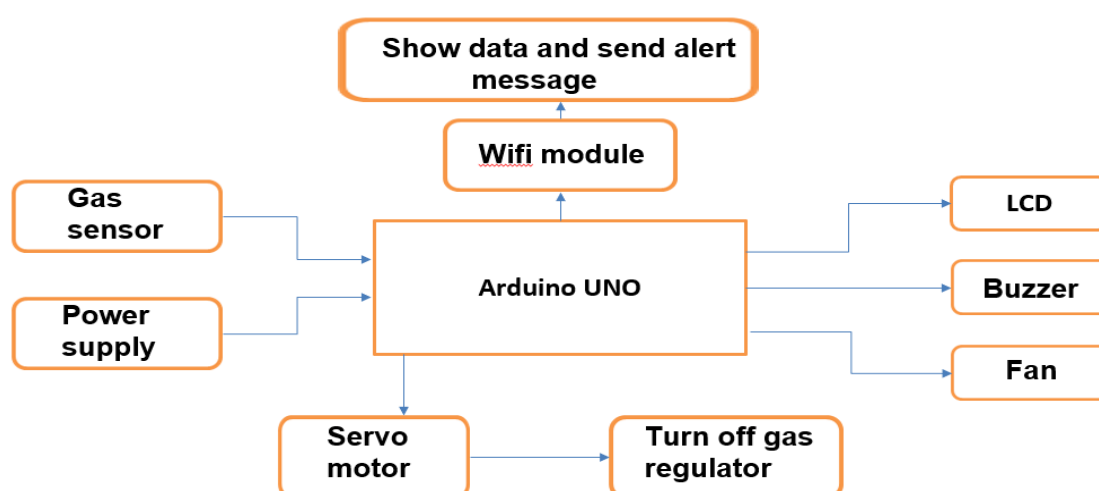


Figure 2.1: Block diagram of LPG gas leakage detection using Arduino

The block diagram represents the overall architecture of the LPG Gas Leakage Detection and Safety Automation System based on the Arduino UNO microcontroller. Each block in the diagram plays a crucial role in achieving real-time detection, alerting, and automatic safety response.

During LPG leakage. The system is designed to sense gas concentration, warn users through multiple channels, activate safety mechanisms, and automatically shut off the gas supply. The following explanation describes the purpose and functioning of each block and how they interact to form a complete safety solution.

The Gas Sensor (MQ-series) is placed at the input of the system and is responsible for continuously detecting the presence of LPG in the surrounding environment. It generates an analogue voltage proportional to the concentration of gas in the air. This signal is sent to the Arduino UNO, which acts as the central processing unit of the system. The Arduino continuously reads the sensor values to determine whether the LPG concentration exceeds the safety threshold. Along with the sensor input, a stable Power Supply is provided to the Arduino as well as all external components, ensuring uninterrupted operation. The power supply block may include adapters, 5V/12V supplies, or batteries, depending on the system design.

The processed output from the sensor is used by the Arduino to trigger various safety responses. When the gas concentration reaches unsafe levels, the Arduino sends signals to different output devices. First among them is the Buzzer, which produces a loud, audible alert to warn people in the surrounding area. This is critical for ensuring that even individuals who are not looking at the system can be alerted immediately. Parallel to the buzzer, the Arduino activates the Fan, typically an exhaust fan, which helps to reduce the gas concentration by improving ventilation. The fan plays a major role in preventing the accumulation of LPG, which could otherwise lead to fire or explosion.

The system also includes an LCD Display, which continuously shows gas concentration levels, system status, and warning messages. This helps users visually monitor the safety condition in real time. The LCD acts as an information interface, allowing the user to know if the system is in safe, warning, or dangerous mode. For remote alerts, a Wi-Fi Module is integrated with the Arduino. This module establishes an internet connection and sends data to a dedicated application, cloud dashboard, or messaging service. When leakage is detected, the Wi-Fi module transmits an alert message to the user's mobile phone, ensuring they are notified even if they are not physically present near the system.

One of the most important components in the block diagram is the Servo Motor, which is responsible for executing the mechanical action of shutting off the gas regulator. When a leakage is detected, the Arduino sends a control signal to the servo motor, which is mechanically linked to the regulator valve. The motor rotates to a predefined angle, effectively turning off the gas supply and preventing further leakage. This component ensures automatic prevention, reducing dependency on human intervention during emergencies. The action of the servo is connected to

the Turn Off Gas Regulator block, which represents the physical closing of the LPG valve.

The Show Data and Send Alert Message block at the top of the diagram represents the system's capability to provide real-time monitoring and remote notifications using the Wi-Fi module. It combines both visual data presentation and alert messaging functions, ensuring that the user receives critical information regardless of their location. This cloud or network-based communication feature enhances safety by enabling timely responses from users or emergency services.

Overall, the block diagram illustrates the seamless interaction between sensing, processing, alerting, controlling, and communicating components. The Arduino UNO operates as the central controller, coordinating all actions based on the gas sensor input. The buzzer and fan provide immediate local safety responses, while the LCD keeps the user informed. The Wi-Fi module ensures remote communication, and the servo motor ensures automatic shut-off of the regulator. Together, these components create a comprehensive LPG detection and safety automation system capable of preventing accidents, protecting property, and saving lives.

2.3 Flowchart

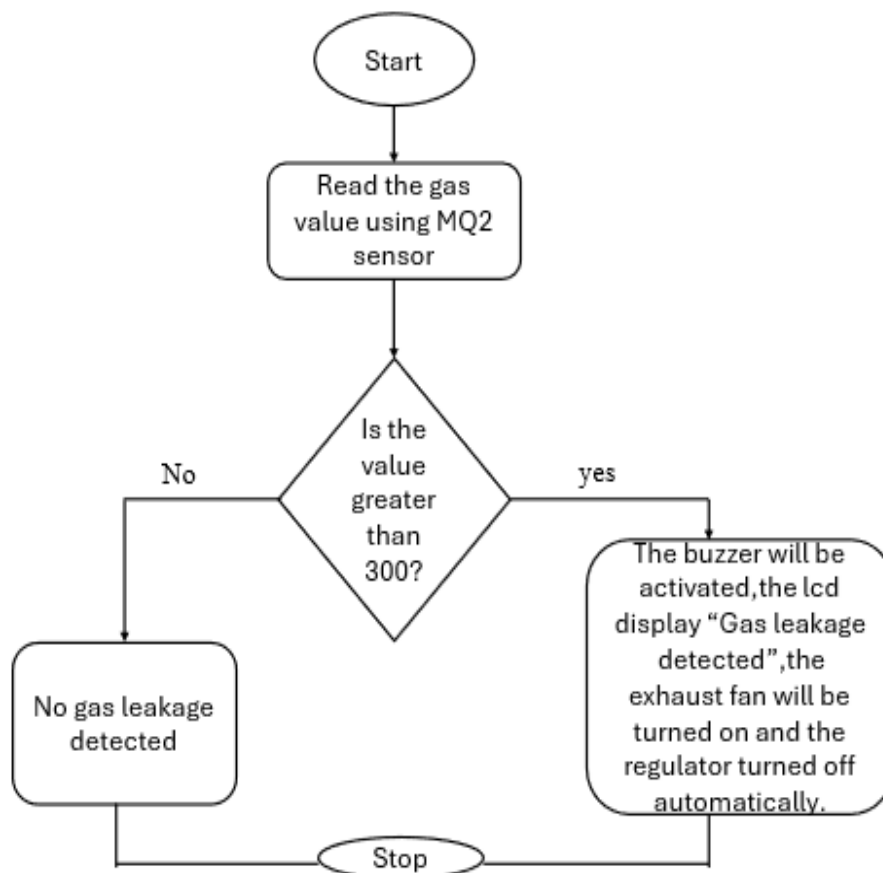


Figure 2.2: Flow chart of the LPG gas detection system

The flowchart represents the complete operational sequence of the Arduino-based LPG Gas Leakage Detection System and clearly explains how the system monitors gas leakage and responds to hazardous situations. The process begins when the system is switched on, marking the Start of the operation. At this stage, all electronic components, such as the MQ2 gas sensor, buzzer, LCD module, exhaust fan, and regulator control mechanism, are initialised by the Arduino. This initialisation ensures that the system is ready to continuously observe the surrounding environment for any possible leakage of LPG gas. Once the system is active, the MQ-2 sensor begins to read the gas concentration in the air.

The MQ-2 sensor is sensitive to LPG, propane, and methane, and it outputs an analogue value that increases when gas concentration rises. The Arduino collects this value and uses it to decide the next step. Continuous reading of gas levels is essential because LPG leaks can happen unexpectedly, and an early warning system is necessary to prevent accidents. After reading the gas value, the flowchart moves to a decision-making stage, where the system checks whether the sensor reading is greater than a predefined threshold value, which in this project is 300. This threshold is set based on sensor calibration and represents the point at which gas concentration becomes unsafe. If the reading is below 300, the system concludes that the environment is safe and there is no gas leakage detected. In this case, no alarms or safety devices are activated. The system simply continues functioning in a normal state and can loop back to the sensor-reading process to maintain continuous monitoring. This helps ensure that the user is not disturbed unnecessarily and the system remains energy-efficient and stable.

However, if the gas sensor reading exceeds 300, the flowchart directs the system to the emergency response path, indicating that a gas leakage has been detected. At this point, the system immediately activates several safety measures designed to alert users and minimise potential hazards. First, the buzzer is activated, producing a loud sound to warn people nearby about the dangerous condition. This immediate audio alert is crucial because it ensures that even if the user is not directly looking at the system, they become aware of the leakage.

Simultaneously, the LCD shows the message "Gas leakage detected", giving a clear visual warning. This message helps the user understand the reason behind the alarm and confirms that the system has detected dangerous gas levels. In addition to warnings, the system also takes preventive measures. The exhaust fan is automatically turned on, which helps to ventilate the area by removing the accumulated gas and allowing fresh air to circulate. Quick ventilation reduces the risk of fire or explosion. Furthermore, the gas regulator is turned off automatically, typically using a servo motor mechanism, to stop additional gas from escaping the cylinder. This is an important safety feature because it prevents the situation from worsening.

After carrying out all these actions, the flowchart reaches the Stop point, marking the completion of this response cycle. Although the flowchart ends here for representation purposes, the actual system would continue monitoring the gas level again to ensure safety at all times. Overall, this flowchart provides a clear and logical representation of how the LPG Gas Leakage Detection System operates. It shows how the system monitors gas concentration, makes decisions based on sensor data, and activates appropriate safety mechanisms during dangerous conditions. The flowchart helps readers understand the step-by-step behaviour of the system and demonstrates how an Arduino-based setup can effectively detect and respond to LPG leakage, making homes and workplaces safer from fire hazards.

2.4 Hardware Requirements

2.4.1 Arduino UNO

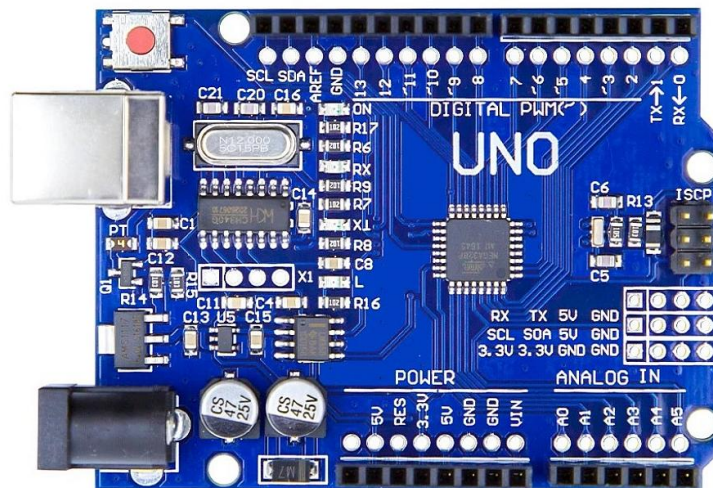


Figure 2.3: Arduino Uno

The Arduino Uno acts as the central processing and control unit of the LPG gas leakage detection system. It continuously monitors the analogue output from the MQ-2 gas sensor, which varies according to the concentration of LPG present in the surrounding air. Using its built-in 10-bit Analogue-to-Digital Converter (ADC), the Arduino converts these analogue signals into digital values, enabling accurate analysis of gas levels. Based on the programmed threshold limits, the microcontroller determines whether the detected gas concentration is safe or dangerous. In the event of gas leakage, the Arduino promptly initiates multiple safety actions. It activates the buzzer to provide an immediate audible alert, turns on the exhaust fan to help disperse the leaked gas, and controls the servo motor to automatically close the gas regulator, thereby preventing further leakage. Simultaneously, the system updates the LCD with real-time status messages such as “Gas Leak Detected” or “System Safe,” ensuring clear visual feedback to users.

Pin	Type	Function
D0 (RX)	Digital I/O	UART Receive
D1 (TX)	Digital I/O	UART Transmit
D2 - D7	Digital I/O	General I/O
D3, D5, D6, D9, D10, D11	PWM	PWM Output
A0 - A5	Analog Input	General I/O, SPI (D10-D13)
GND	Power	Ground
3.3V	Power	3.3V Output
5V	Power	5V Output
Vin	Power	External Power Input
RESET	Control	ADC Reference

Table 2.1: Arduino Uno Pin Description

2.4.2 MQ-2 Gas Sensor



Figure 2.4: MQ-2 gas sensor

The MQ-2 gas sensor is commonly used for detecting combustible gases such as LPG, methane, propane, butane, and smoke. It operates based on the change in resistance of its sensing element when exposed to gas. The sensor uses tin dioxide (SnO_2) as the sensitive material, which has low conductivity in clean air and higher conductivity in the presence of combustible gases. The MQ-2 provides an analogue output that varies according to gas concentration and can also give a digital output through an onboard comparator. It works with a 5V power supply, making it compatible with Arduino microcontrollers. The sensor requires a short preheating time to stabilise and produce accurate readings. Due to its fast response, low cost, adjustable sensitivity, and ease of interfacing, the MQ-2 sensor is widely used in LPG gas leakage detection and safety systems.

2.4.3 GSM Module (SIM800L)



Figure 2.5: GSM module

The GSM module is used in the LPG gas leakage detection system to provide reliable wireless communication and alert functionality. It allows the system to send SMS notifications to predefined mobile numbers when a gas leak is detected, ensuring quick user awareness even when no one is present at the site. The module operates using a SIM card and works on standard GSM cellular networks, similar to a mobile phone. It communicates with the Arduino through serial (UART) communication and is controlled using AT commands. The GSM module requires a stable power supply for proper operation and does not depend on internet connectivity. Due to its ability to provide real-time alerts, ease of integration, and wide network availability, the GSM module significantly enhances the safety and effectiveness of LPG gas leakage detection systems.

2.4.4 Buzzer



Figure 2.6: Buzzer

The buzzer is an important audio alert device used in the LPG gas leakage detection system to warn users about hazardous conditions. When the gas concentration exceeds the safe limit, the Arduino activates the buzzer to produce a loud and continuous sound, drawing immediate attention to the leakage. The buzzer is connected to a digital output pin of the Arduino and operates on a low DC voltage, usually 5V. It provides a fast response, ensuring timely alerts during emergencies. Both active and passive buzzers can be used, with active buzzers generating sound automatically

when powered. Due to its low power consumption, low cost, and reliable operation, the buzzer plays a crucial role in enhancing safety by providing instant audible warnings in LPG gas leakage detection systems.

2.4.5 Exhaust Fan



Figure 2.7: Exhaust Fan

The exhaust fan is a vital safety component in the LPG gas leakage detection system, used to reduce the concentration of leaked gas in the surrounding area. When the Arduino detects a gas level above the safe threshold, it automatically activates the exhaust fan to improve ventilation. The fan helps quickly disperse accumulated LPG, thereby reducing the risk of fire or explosion. It is usually driven through a relay or motor driver since it operates at a higher current or voltage than the Arduino can supply directly. The exhaust fan ensures continuous air circulation until the gas level returns to a safe range. By rapidly removing hazardous gases from the environment, the exhaust fan significantly enhances the effectiveness and reliability of the LPG gas leakage detection and safety system.

2.4.6 LCD Display

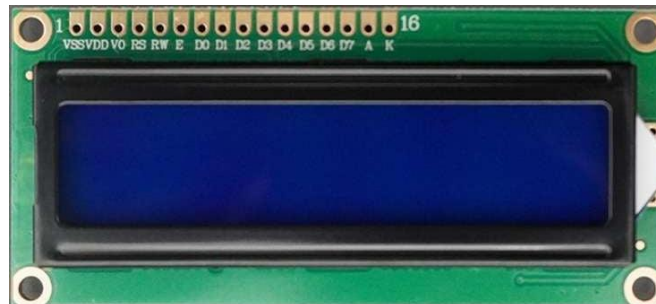


Figure 2.8: LCD Display

The LCD is used in the LPG gas leakage detection system to provide clear visual information about the system status. It displays messages such as “Gas Leak Detected,” “System

Safe,” and real-time sensor readings, allowing users to easily understand the condition of the environment. The LCD is interfaced with the Arduino using parallel communication or I²C protocol, which reduces the number of connecting wires. It operates on a 5V power supply, making it compatible with Arduino-based systems. The display updates continuously based on sensor data and system actions. Due to its low power consumption, good readability, and ease of interfacing, the LCD enhances user interaction and monitoring in the LPG gas leakage detection system.

2.4.7 Servo Motor for Gas Regulator



Figure 2.9: Servo motor

A servo motor is a precision-controlled motor widely used in automation and embedded systems for accurate position control. It operates using a closed-loop feedback system, where the motor continuously adjusts its position based on control signals. Servo motors are commonly controlled using Pulse Width Modulation (PWM) signals generated by microcontrollers like Arduino. The motor shaft rotates to a specific angle, usually between 0° and 180°, depending on the pulse width of the signal. Servo motors consist of a DC motor, gear mechanism, control circuit, and position sensor (potentiometer). They are known for high accuracy, quick response, and stable operation. In LPG gas leakage detection systems, servo motors are often used to automatically close the gas regulator during leakage. This automatic shut-off mechanism enhances safety by preventing further gas flow. Servo motors consume power only when movement or holding torque is required. Their compact size and easy interfacing make them suitable for low-cost safety and robotic applications. Overall, servo motors play a crucial role in systems requiring controlled and reliable mechanical movement.

2.5 Software Requirements

2.5.1 Arduino IDE



Figure 2.10: Arduino IDE

Arduino IDE (Integrated Development Environment) is an open-source software. It is a platform which can be used for applications on different OS as Windows, macOS, Linux is making a cross-platform. It includes a code editor with features such as syntax highlighting, brace matching, and automatic indentation, and is also capable of compiling and uploading programs to the board with a single click. Arduino IDE uses functions from C++ and C for programming. Arduino IDE is used for writing the code, as it supplies a software library and uploads it to the Arduino-compatible boards. 2.3.2 Arduino Bluetooth Controller.

2.6 Summary

The chapter explains the design and implementation of an Arduino-based LPG Gas Leakage Detection System. It describes the block diagram, which includes the MQ-2 gas sensor, Arduino controller, buzzer, exhaust fan, GSM/Wi-Fi module, LCD, and servo motor. The MQ-2 sensor continuously monitors LPG levels and sends signals to the Arduino, which processes the data to detect unsafe gas concentration. When leakage is detected, the buzzer provides an immediate audible warning, while the exhaust fan turns on to reduce gas accumulation. The LCD's real-time sensor values and alert messages ensure users stay informed. A GSM or Wi-Fi module sends notifications to the user for remote monitoring. The servo motor automatically closes the gas regulator to prevent further leakage. The chapter also explains the hardware and software requirements and summarises how all components work together to create a reliable, safe, and automated LPG detection system.

Results and Discussion

3.1 System Performance

The system performance of the LPG Gas Leakage Detection System using Arduino is efficient, reliable, and fast in detecting harmful gas leaks. The MQ gas sensor continuously monitors the surrounding air and responds quickly when LPG concentration crosses the safety threshold. The Arduino processes the sensor readings accurately and activates safety mechanisms within seconds. The buzzer produces an immediate audible alert to warn people nearby, while the exhaust fan automatically turns on to remove accumulated gas from the area. The LCD clearly shows real-time gas levels and system status. The GSM module enhances performance by sending instant SMS alerts to the user's mobile phone, even when they are away. The system operates with low power consumption and works continuously without manual intervention. Its components perform consistently under normal conditions, offering stable and dependable operation. Automatic shut-off of the gas regulator further increases safety by stopping additional leakage. The design is simple, cost-effective, and suitable for home or industrial use. Overall, the system provides fast detection, quick response, and effective safety actions, ensuring maximum protection against LPG leakage hazards.

3.2 Functionality Testing

Functionality testing of the LPG Gas Leakage Detection System ensures that every component operates correctly and performs its intended action during gas leakage conditions. The MQ gas sensor was tested by exposing it to small amounts of LPG, and the Arduino successfully detected the rise in gas concentration above the preset threshold. Once gas was detected, the buzzer activated immediately, providing a loud alarm, and the exhaust fan turned ON automatically to ventilate the area. The LCD displayed the correct status messages such as "Gas Detected" and "Safety Activated." The GSM module was tested by sending alert messages to the registered mobile number, and it consistently delivered SMS notifications within a few seconds. The automatic gas regulator shut-off mechanism also responded correctly by closing the valve when leakage levels were high. Each component sensor, buzzer, fan, GSM, LCD, and servo was individually tested and then tested together as a full system to ensure proper coordination. The system functioned reliably under repeated trials, showing stable performance and quick response. Overall, functionality testing confirmed that the system accurately detects gas leakage and

activates all safety measures effectively.

3.3 Future Scope

- 1. IoT Integration**

The system can be connected to the Internet (using Wi-Fi/GSM) to send real-time alerts to mobile phones or cloud platforms.

- 2. Mobile App Monitoring**

A dedicated Android/iOS app can be developed to monitor gas levels, alerts, and system status remotely.

- 3. Automatic Gas Shut-off System**

An electronic gas valve can be added to automatically stop gas supply during leakage.

- 4. SMS and Call Alerts**

GSM modules can be used to send SMS or make emergency calls to users and authorities.

- 5. Voice Alerts and Multilingual Support**

Voice warnings in multiple languages can be added for better awareness.

- 6. Data Logging and Analysis**

Gas concentration data can be stored and analysed to predict leakage patterns and improve safety.

- 7. Integration with Smart Home Systems**

The detector can be integrated with smart home platforms like Alexa or Google Home.

- 8. Improved Sensors and Accuracy**

Advanced gas sensors can be used for higher sensitivity, faster response, and lower false alarms.

- 9. Battery Backup and Solar Power**

Adding backup power ensures continuous operation during power failures.

- 10. Industrial and Commercial Applications**

The system can be scaled for use in factories, gas plants, hotels, and commercial kitchens.

3.4 Display output

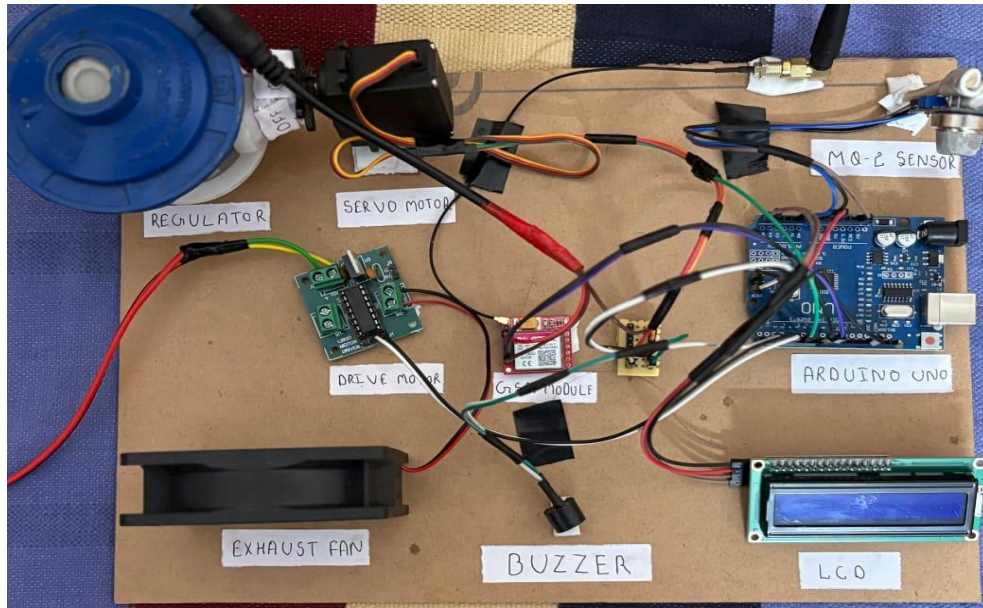


Figure 3.1: Result When the circuit is Off

The image shows the assembled prototype of the LPG Gas Leakage Detection System using Arduino. The setup includes an Arduino Uno, an MQ gas sensor, a GSM module, an LCD, an exhaust fan, and a servo motor connected to a gas regulator model. All components are interconnected using jumper wires on a mounting board. The Arduino controls the sensor readings and activates the fan, buzzer, LCD, and regulator shut-off during gas leakage. The GSM module is included to send alert messages to the user. Overall, the image represents the system when the circuit is off.

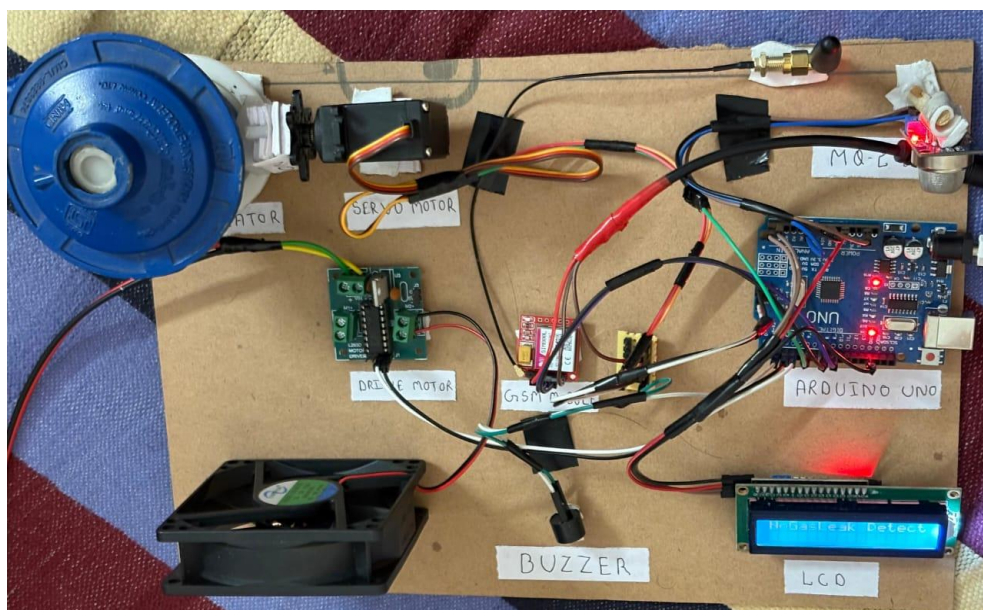


Figure 3.2: Result when the circuit is On



Figure 3.3: Screenshot of the Message

The above image shows the screenshot of the message “Gas Leak detected” sent from the GSM module to the owner.

Conclusion

The LPG Gas Leakage Detection System using Arduino successfully demonstrates an effective and reliable approach to enhancing safety in residential and industrial environments. The project integrates multiple components such as the MQ gas sensor, Arduino microcontroller, GSM module, exhaust fan, buzzer, LCD, and an automatic regulator shut-off mechanism to create a complete and responsive safety unit. Throughout the development and testing stages, the system proved capable of accurately detecting even small concentrations of LPG in the air and immediately activating essential safety measures. The fast response time of the buzzer and exhaust fan helps reduce the accumulation of leaked gas, while the automatic valve shut-off minimises the chances of further leakage.

One of the most important features of the system is its ability to send SMS alerts through the GSM module, allowing users to take timely action even when they are not physically present near the system. This remote alert capability significantly increases the practical value and real-world usefulness of the project. The LCD further supports real-time monitoring by showing the current gas level and system status, making the setup user-friendly and informative. Despite minor challenges such as sensor calibration, wiring complexity, and network dependency for message delivery, the system overall performed efficiently and reliably under repeated testing conditions.

This project highlights the importance of automation and early detection in preventing accidents related to LPG leakage. It also demonstrates how embedded systems and low-cost microcontrollers like Arduino can be used to build practical and life-saving applications with high effectiveness. The system is economical, easy to install, and can be further enhanced with IoT technologies, mobile apps, and cloud monitoring in future versions.

In conclusion, the LPG Gas Leakage Detection System using Arduino successfully achieves its goal of providing a safe, automated, and efficient method for detecting gas leakage and alerting users. The project contributes significantly to household and industrial safety, showcasing a practical solution that combines technology, innovation, and preventive action to protect lives and property.

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Appendix

```
#include <wire.h>
#include <SoftwareSerial.h>
SoftwareSerial myser (8,9);

#include <Servo.h>
Servo myservo;
#include <LiquidCrystal I2C.h>
LiquidCrystal_I2C lcd (0x27,16,2);
int gas = A0;
int buzz = 2;
int f1 = 3;
int f2 = 4;
void setup ( ) {          // put your setup code here, to run once:
Serial. begin (9600);
myser. begin (9600);
pinMode (buzz, OUTPUT);
pinMode (f1, OUTPUT);
pinMode (f2, OUTPUT);
myservo. attach (6);
myservo. write (0);
lcd.init ( );
lcd.backlight ( );
lcd.clear ( );
}

Void loop ( ) {          // put your main code here, to run repeatedly:
Int gasv = analogRead (gas);
Serial. println (gasv);
If (gasv>300) {
digitalWrite (f1,1);
digitalWrite (f2,0);
digitalWrite (buzz,1);
```

```

myservo. write (90);
lcd.clear ( );
lcd.print ("GasLeak Detected");
lcd.setCursor (0,1);
Icd.print ("Value:"+String(gasv));
Delay (2000);
digitalWrite (buzz,0);
msg1 ( );
else
digitalWrite (f1,0);
digitalWrite (f2,0);
lcd.clear ( );
lcd.print ("NoGasLeak Detected");
myservo. write (0);
delay (1000);
void msg1 ( ){
myser.println ("AT+CMGF=1");    //Because we want to send the SMS in text mode
delay (2000);
myser.println ("AT+CMGS=\"8296446188\"");    // change to the phone number you using
delay (2000);
myser.print ("Gas Leak Detected");
delay (2000);
myser. write (0x1A);
delay (2000)

```